

A 6-Week Transition to Maximal Running Shoes Does Not Change Running Biomechanics

J.J. Hannigan,^{*†} PhD, ATC, CSCS, and Christine D. Pollard,[†] PhD, PT

Investigation performed at the FORCE Laboratory, Oregon State University–Cascades, Bend, Oregon, USA

Background: A recent study suggested that maximal running shoes may increase the impact force and loading rate of the vertical ground-reaction force during running. It is currently unknown whether runners will adapt to decrease the impact force and loading rate over time.

Purpose: To compare the vertical ground-reaction force and ankle kinematics between maximal and traditional shoes before and after a 6-week acclimation period to the maximal shoe.

Study Design: Controlled laboratory study.

Methods: Participants ran in a traditional running shoe and a maximal running shoe during 2 testing sessions 6 weeks apart. During each session, 3-dimensional kinematics and kinetics were collected during overground running. Variables of interest included the loading rate, impact peak, and active peak of the vertical ground-reaction force, as well as eversion and dorsiflexion kinematics. Two-way repeated measures analyses of variance compared data within participants.

Results: No significant differences were observed in any biomechanical variable between time points. The loading rate and impact peak were higher in the maximal shoe. Runners were still everted at toe-off and landed with less dorsiflexion, on average, in the maximal shoe.

Conclusion: Greater loading rates and impact forces were previously found in maximal running shoes, which may indicate an increased risk of injury. The eversion mechanics observed in the maximal shoes may also increase the risk of injury. A 6-week transition to maximal shoes did not significantly change any of these measures.

Clinical Relevance: Maximal running shoes are becoming very popular and may be considered a treatment option for some injuries. The biomechanical results of this study do not support the use of maximal running shoes. However, the effect of these shoes on pain and injury rates is unknown.

Keywords: maximal running shoes; impact force; loading rate; eversion

Injury rates among runners remain high,^{24,29} despite significant advances in running shoe technology over the past 50 years. Traditional running shoes generally contained a moderate amount of cushioning, 8- to 12-mm of

heel-toe drop, and varied per the amount of rearfoot control that they offered.⁹ Over the past 10 years, however, companies began significantly varying the amount of cushioning in the running shoe, with both minimally and maximally cushioned shoes being marketed and sold.

Since 2010, maximally cushioned running shoes have become increasingly popular, especially among recreational runners. There is currently no academic definition of a maximal running shoe, but the defining industry feature is substantially greater cushioning across the midsole as compared to a traditional running shoe. Many variations of this shoe design exist; however, the exact stack height, amount of heel-toe drop, presence of a heel or forefoot rocker, and type of midsole foam are among several factors that differentiate the maximal shoes currently offered by shoe companies.

Very little research has been performed on maximal running shoes. A previous study found that both before and after a 5-km run on the treadmill, the average vertical loading rate (AVLR) and vertical impact peak (VIP) of the

^{*}Address correspondence to J.J. Hannigan, PhD, ATC, CSCS, Program in Kinesiology, Oregon State University–Cascades, 1500 SW Chandler Avenue, Bend, OR 97702, USA (email: jj.hannigan@osucascades.edu).

[†]Program in Kinesiology, Oregon State University–Cascades, Bend, Oregon, USA.

The authors declared that they have no conflicts of interest in the authorship and publication of this contribution. AOSSM checks author disclosures against the Open Payments Database (OPD). AOSSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

vertical ground-reaction force (vGRF) were higher in the maximal shoe as compared to a traditional shoe during overground running,²¹ which may increase the risk of injury.^{7,10,15,19,31} However, other research found no differences in vGRF measures between maximal and traditional shoes^{1,2} or that differences existed only during downhill running.⁸ These conflicting results could be related to the specific maximal and traditional shoes used in each study.

To our knowledge, almost all research on maximal running shoes to date has been on the immediate effect of the shoes during a single testing session.^{2,8,21,25,26} While 1 study examined the effect of a 4-week transition to maximal running shoes, this study did not report the effect of time on ground-reaction forces or ankle kinematics.¹ It is currently unknown whether a longer acclimation period would affect these biomechanical variables. Thus, the primary purpose of the study was to examine the effect of a 6-week transition to maximal footwear on running biomechanics (ground-reaction forces and ankle kinematics) and to compare running biomechanics between maximal and traditional running shoes. It was hypothesized that the AVL and VIP of the vGRF would be significantly higher initially in the maximal shoe but that these variables would significantly decrease at the 6-week follow-up, negating any differences between shoes. No differences between shoes or changes over time were hypothesized for ankle kinematics.

METHODS

Participants

Participants were included in this study if they were 18 to 45 years old, ran at least 15 miles per week, had not suffered a major injury within the past 6 months, and had never trained in a minimal or maximal running shoe. A major injury was defined as pain that caused a restriction or stoppage of running for at least 7 days or 3 consecutive scheduled training sessions or that required the runner to consult a physician.³⁰ On the basis of these criteria, 28 runners were included in the study. All participants signed an informed consent document approved by the university's institutional review board before participating in the first day of testing.

Instrumentation

Running kinematics were collected with a Vicon 8-camera motion analysis system (Oxford Metrics LTD) at a sampling frequency of 250 Hz. Ground-reaction forces were collected from 2 embedded force platforms sampling at 1000 Hz (Advanced Mechanical Technology, Inc). Kinematic and kinetic data were synchronized and collected with Vicon Nexus software (v 1.8.5).

Procedures

Participants came to the biomechanics laboratory for 2 separate biomechanical testing sessions, separated by approximately 6 weeks. For both testing sessions, participants wore



Figure 1. The maximal shoe (Hoka One One Bondi 5; left) and traditional shoe (New Balance 880v2; right) used in this study.

both a traditional running shoe (New Balance 880v2: drop, 10 mm; heel height, 33 mm; forefoot height, 23 mm) and a maximal shoe (Hoka One One Bondi 5: drop, 4 mm; heel height, 33 mm; forefoot height, 29 mm) (Figure 1). Participants were fitted for the maximal shoe at a local running shoe retailer before participation and for the traditional shoe during their first study visit. Shoe order was randomized for each participant at each testing session.

Testing procedures were the same for both testing sessions. Reflective markers (14-mm spheres) were placed bilaterally over the following anatomic landmarks: the first and fifth metatarsal heads, distal interphalangeal joint of the second toe, medial and lateral malleoli, medial and lateral femoral epicondyles, greater trochanters, iliac crests, anterior superior iliac spines, and the joint space between the fifth lumbar and first sacral spinous processes. The foot markers were placed directly on the shoe. Rigid clusters of 4 reflective markers were attached bilaterally to the participant's thigh and leg with a custom adhesive taping system. In addition, rigid clusters of 3 reflective markers were placed bilaterally on the heel counter of the shoe. Markers were always placed by the same researcher (J.J.H.), who had several years of experience placing markers. Following marker placement, the participant was asked to stand in the center of the calibration area to collect a static calibration trial. Participants then changed into the other shoe, and the calibration process was repeated. Both calibration trials occurred before anatomic markers were removed. Once the calibration trial was captured, all markers were removed except the clusters and the iliac crest, anterior superior iliac spine, and sacral markers.

The participants were allowed 3 to 5 practice trials to become familiar with the procedures and instrumentation. The participants then completed 5 successful running trials for the dominant leg (defined as the leg that they prefer to kick a soccer ball with). For each trial, participants ran approximately 10 m before the capture volume and continued running for 2 to 3 strides after the force plates. All participants were instructed to run at a pace of approximately 8 minutes per mile, which was determined with timing gates within the capture volume. Trials were considered successful only with a clean foot strike on the force plate and within an

TABLE 1
Mileage Schedule

Week	Maximal Shoe Mileage, % ^a
1	20
2	40
3	60
4	80
5	100
6	100

^aThe percentage of the runner's normal mileage run in the maximal shoe.

acceptable running pace window ($\pm 5\%$ from 8-minute mile). After completion of the 5 successful running trials, participants switched shoes, and the process was repeated.

Participants were given a maximal shoe transition schedule at the end of the first visit. Participants were instructed to maintain their normal mileage while gradually increasing the percentage of mileage in the maximal shoes each week. Table 1 shows the percentage of total mileage run in the maximal shoe each week.

Data Analysis

Coordinate data were digitized in Vicon Nexus software. Kinematic data were filtered with a fourth-order zero-lag Butterworth 12-Hz low-pass filter, while kinetic data were filtered with a fourth-order zero-lag Butterworth 50-Hz low-pass filter.²³ Visual3D software (v 5; C-Motion, Inc) was used to quantify 3-dimensional ankle joint kinematics. Joint kinematics were calculated with a joint coordinate system approach.¹² The peak angle for eversion and dorsiflexion was defined as the maximum angle during the stance phase, while excursion was defined as the difference between the angle at initial contact and the peak angle. The AVLr was calculated over the middle 60% of the vGRF curve from heel strike to the VIP, consistent with Milner et al.¹⁵ These calculations were made with custom Excel software.

Statistical Analysis

Variables of interest included the AVLr, VIP, and vertical active peak (VAP) of the vGRF (normalized by body weight), as well as eversion and dorsiflexion kinematics (angle at initial contact, peak angle, excursion, angle at toe-off). Differences were determined by 2-way repeated measures analyses of variance (shoe $\times$ time), with the omnibus alpha level set to .05. Post hoc comparisons were made with appropriate Bonferroni corrections. All statistical testing was performed with SPSS (v 25; IBM), except for effect size (ES; Cohen's *d*), which was calculated with G*Power (v 3.1; G*Power).

RESULTS

Three female participants dropped out of the study between the initial and 6-week study visits for blisters,

TABLE 2
Participant Characteristics

Variable	Mean $\pm$ SD
Women:men, n	14:6
Age, y	34.6 $\pm$ 7.3
Height, m	1.72 $\pm$ 9.9
Body mass, kg	65.8 $\pm$ 11.3
Weekly mileage	22.1 $\pm$ 4.8
Running experience, y	10.5 $\pm$ 6.8

medial shin pain, and Achilles pain at 2, 3, and 5 weeks, respectively. In addition, on the basis of visual inspection of the vGRF data and 2-dimensional video, 5 participants (3 women, 2 men) were determined to have a non-rearfoot strike pattern in one or both shoes. These 8 participants were not included in this analysis. Table 2 presents the participant characteristics for the 20 runners included in this analysis.

No shoe $\times$ time interactions were observed for AVLr or VIP. There was a significant main effect of shoe for AVLr and VIP, as both AVLr ($P < .001$, ES = 1.37) and VIP ($P < .001$, ES = 1.70) were significantly higher in the maximal shoe as compared to the traditional shoe (Table 3). No main effect of time was observed for AVLr or VIP. There was a significant shoe $\times$ time interaction effect for VAP, as it significantly decreased in the traditional shoe between visits ($P = .001$, ES = 0.59) (Table 3). No main effect of shoe was observed for VAP ($P = .917$, ES = 0.00) (Table 3).

No interaction effects or main effects of time were observed for any kinematic variable. There were significant main effects of shoe for eversion kinematics. While peak eversion was not different between shoes ($P = .313$, ES = 0.16), runners landed with significantly less inversion ($P < .001$, ES = 1.36) and had significantly less eversion excursion ($P < .001$, ES = 1.84) in the maximal shoe (Figure 2, Table 3). In addition, we saw significant differences in the eversion angle at toe-off ($P < .001$, ES = 1.17); on average, runners were still everted at toe-off in the maximal shoe while inverted at toe-off in the traditional shoe.

There were also significant main effects of shoe for dorsiflexion kinematics. The dorsiflexion angle at initial contact ($P = .002$, ES = 0.62) and the peak dorsiflexion angle ($P = .002$, ES = 0.67) were significantly greater in the traditional shoe as compared to the maximal shoe (Table 3). No differences were observed for angular excursion ($P = .293$, ES = 0.17).

DISCUSSION

The primary purposes of this study were to examine whether a 6-week acclimation period to maximal running shoes significantly changed running biomechanics (ground-reaction forces and ankle kinematics) and to compare running biomechanics between maximal and traditional footwear. While differences in ground-reaction forces and ankle kinematics were observed between shoes,

TABLE 3
Biomechanical Variables of Interest^a

	Initial Visit		Second Visit		Average	
	Traditional	Maximal	Traditional	Maximal	Traditional	Maximal
Ground-reaction forces						
Loading rate, BW/s	69.0 ± 16.7	85.9 ± 20.1	66.3 ± 12.6	83.2 ± 17.5	67.6 ± 14.7	84.5 ± 18.7 ^b
Impact peak, BW	1.69 ± 0.20	1.98 ± 0.21	1.69 ± 0.26	1.97 ± 0.28	1.69 ± 0.23	1.98 ± 0.13 ^b
Active peak, BW	2.43 ± 0.12	2.40 ± 0.12	2.39 ± 0.13 ^c	2.42 ± 0.13	2.41 ± 0.13	2.41 ± 0.13
Eversion kinematics, ^d deg						
Angle at initial contact	3.54 ± 3.60	0.43 ± 3.36	3.61 ± 3.15	0.81 ± 3.01	3.58 ± 3.34	0.62 ± 3.16 ^b
Peak angle	-10.6 ± 3.06	-9.64 ± 2.74	-10.4 ± 2.61	-10.7 ± 2.73	-10.5 ± 2.81	-10.2 ± 2.75
Angular excursion	14.1 ± 2.38	10.1 ± 2.53	14.0 ± 2.85	11.50 ± 2.95	14.1 ± 2.59	10.8 ± 2.81 ^b
Angle at toe-off	1.39 ± 3.1	-1.05 ± 3.56	1.13 ± 4.52	-2.1 ± 4.21	1.26 ± 3.82	-1.58 ± 3.88 ^b
Dorsiflexion kinematics, deg						
Angle at initial contact	8.83 ± 3.90	6.68 ± 3.17	9.1 ± 3.47	6.46 ± 5.35	8.97 ± 3.65	6.57 ± 4.35 ^b
Peak angle	23.3 ± 2.43	22.1 ± 3.64	24.5 ± 1.90	22.1 ± 2.64	23.9 ± 2.23	22.1 ± 3.14 ^b
Angular excursion	14.5 ± 3.09	15.4 ± 2.70	15.4 ± 3.68	15.6 ± 4.58	14.9 ± 3.39	15.5 ± 3.71

^aValues are presented as mean ± SD. BW, body weight.

^bRepresents a main effect of shoe, $P < .05$.

^cRepresents an interaction effect (shoe × time), $P < .05$.

^dFor eversion kinematics results, eversion is negative, while inversion is positive.

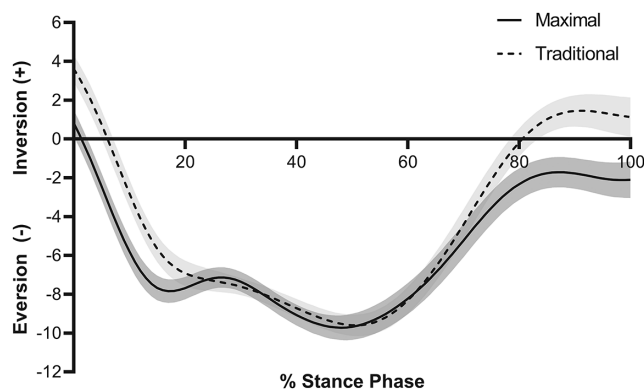


Figure 2. Ensemble curves for inversion (+) and eversion (–) in each shoe condition, in degrees. Error bars represent ± 1 SE.

no changes over time were observed (for either overall main effects or effects for only the maximal shoe). These findings did not support our initial hypothesis.

A 6-week acclimation to minimal shoes was previously found to decrease the VIP and AVLR of the vGRF in habitually shod runners.¹⁴ On the basis of this finding, we hypothesized that the same phenomenon may occur in maximal shoes. However, no significant changes in ground-reaction forces or ankle kinematics were observed in maximal shoes after 6 weeks. All runners included in this analysis successfully transitioned to running 100% of their mileage in the maximal shoe for the final 2 weeks of the 6-week acclimation period. During a 4-week acclimation study where runners exclusively wore maximal shoes, Agresta et al¹ also found no biomechanical changes over time in foot strike angle, spatiotemporal parameters, leg

accelerations, or leg stiffness. It is possible that an acclimation period longer than 4 to 6 weeks is required to see biomechanical changes during running, such as a reduction in vGRF parameters. It is also possible that the extra cushioning in the maximal shoe may decrease proprioceptive feedback from the foot during running,^{18,22} inhibiting a runner's ability to make biomechanical adjustments over time. A minimal shoe likely offers much more proprioceptive feedback,^{18,22} enhancing a runner's ability to make appropriate biomechanical adjustments.

While no changes were seen over time, we did see greater AVLR and VIP in the maximal shoes, which supports the findings from a previous study with the same shoes.²¹ There is currently debate on whether the VIP or AVLRs are strong predictors of injury,^{16,17} as external forces may not be highly correlated with the internal forces acting on musculoskeletal structures.¹⁷ However, several studies found a relationship between high VIP or AVLRs and injury,^{7,10,15,19,31} with some of this evidence being prospective.^{7,10} Thus, the higher AVLRs and impact forces observed during this study may suggest an increased risk of injury in these maximal shoes. It is important to note that 2 runners did drop out of the study during the 6-week transition to maximal shoes owing to musculoskeletal pain, but this cannot necessarily be attributed to higher VIPs or AVLRs.

Other studies did not find differences in the AVLR or VIP between maximal and traditional running shoes.^{1,2,8} This may be attributed to the fact that these studies used different maximal and traditional shoes. Also, while we required all participants to run at the same pace (8 min/mile), other studies had participants run at a self-selected pace.^{1,2,8} The rockered midsole design of the maximal shoe is also noteworthy: while a rockered shoe has not been shown to affect vGRF parameters,⁶ more evidence is needed in this area. Regardless, since

the AVLr and VIP found in the current study were drastically higher than previous studies,^{2,8} we recommend that researchers not consider all maximal shoes the same.

While no differences over time were observed for eversion or dorsiflexion kinematics, we did see differences between shoes for several variables. On average, runners landed with greater inversion in the traditional shoe, had greater eversion excursion in the traditional shoe, and toed off with greater inversion in the traditional shoe. No differences were seen in peak eversion, which has been associated with an increase in injury risk.^{5,20} However, the differences observed in eversion angle at toe-off could be clinically relevant.

During running, inversion of the foot before toe-off locks the transverse tarsal joint and creates a rigid lever for push-off.¹³ If this does not occur, the tissues of the lower leg may experience greater strain.^{4,13} Greater eversion duration during running was associated with an increased risk of developing Achilles tendinopathy and medial tibial stress syndrome.^{4,5} While we did not directly quantify duration of eversion in this study, runners in this study, on average, were still everted at toe-off in the maximal shoe. This finding may be attributed to a lack of stability in highly cushioned shoes and could help explain why 2 runners dropped out of the study due to Achilles pain and medial shin pain.

Runners landed with greater dorsiflexion in the traditional shoe and had greater peak dorsiflexion in the traditional shoe as compared to the maximal shoe. The maximal shoe used in this study has a rocker midsole design, which was found to affect dorsiflexion kinematics during running.^{6,27,28} While these previous results are mixed, it is possible that the rocker design was partially responsible for the differences that we saw in dorsiflexion kinematics. A greater heel-toe drop, as seen in the traditional shoe, encourages more of a heel strike⁹ and may also help explain our results. It is interesting to note that impact forces were still significantly lower in the traditional shoe despite landing in a more dorsiflexed position.

Several limitations exist for this study. First, all runners kept their normal training volume, meaning that there was some variability in the absolute mileage run in the maximal shoes during the 6-week transition across participants. We investigated only rearfoot strikers in this analysis, so we cannot make any conclusions about the effect of maximal shoes over time in habitual midfoot or forefoot strikers. We do not have data on the mechanical properties of either shoe, which can affect running biomechanics.³ Placing markers directly on the shoe limits our ability to accurately quantify ankle kinematics; however, the longitudinal nature of this study precluded cutting holes in the shoe. Women and men were analyzed together in this study, despite cited differences in running biomechanics between sexes.¹¹ We initially included sex as a between-participants factor in our analysis but found no main effect of sex and no interaction effects between sex and shoe, or sex and time. Thus, to simplify the reporting of our results, we pooled men and women together. Finally, our study tested only 1 model of maximal shoe, meaning that our results may or may not be applicable to other maximal shoe models.

CONCLUSION

A 6-week acclimation to maximal running shoes did not change running biomechanics, as the AVLr and VIP of the vGRF were still elevated as compared to a traditional shoe after this period. In addition, differences in ankle kinematics were observed; most notably, greater eversion at toe-off was seen in the maximal shoe. Each of these factors may increase a runner's risk of injury. If runners do decide to try a maximal shoe, the results of this study indicate that they should exercise caution during the transition. In addition, clinicians may want to avoid recommending maximal running shoes until more research has been performed.

ACKNOWLEDGMENTS

The authors acknowledge Jacqueline Diulio, Brian Newcomb, Sherri Dean, Bella Krevitz, and Justin Ter Har for their assistance in data collection and analysis.

REFERENCES

1. Agresta C, Kessler S, Southern E, Goulet GC, Zernicke R, Zendler JD. Immediate and short-term adaptations to maximalist and minimalist running shoes. *Footwear Sci*. 2018;4280:1-13.
2. Aminaka N, Arthur K, Porcari JP, Foster C, Cress M, Hahn C. No immediate effects of highly cushioned shoes on basic running biomechanics. *Kinesiology*. 2018;50(1):124-130.
3. Baltich J, Maurer C, Nigg BM. Increased vertical impact forces and altered running mechanics with softer midsole shoes. *PLoS One*. 2015;10(4):1-11.
4. Becker J, James S, Wayne R, Osternig L, Chou L-SS. Biomechanical factors associated with Achilles tendinopathy and medial tibial stress syndrome in runners. *Am J Sports Med*. 2017;45(11):2614-2621.
5. Becker J, Nakajima M, Wu WFW. Factors contributing to medial tibial stress syndrome in runners. *Med Sci Sport Exerc*. 2018;50(10):2092-2100.
6. Boyer KA, Andriacchi TP. Changes in running kinematics and kinetics in response to a rocker shoe intervention. *Clin Biomech*. 2009;24(10):872-876.
7. Bredeweg SW, Kluitenberg B, Bessem B, Buist I. Differences in kinetic variables between injured and noninjured novice runners: a prospective cohort study. *J Sci Med Sport*. 2013;16(3):205-210.
8. Chan ZYS, Au IPH, Lau FOY, Ching ECK, Zhang JH, Cheung RTH. Does maximalist footwear lower impact loading during level ground and downhill running? *Eur J Sport Sci*. 2018;18(8):1083-1089.
9. Davis IS. The re-emergence of the minimal running shoe. *J Orthop Sport Phys Ther*. 2014;44(10):775-784.
10. Davis IS, Bowser BJ, Mullineaux DR. Greater vertical impact loading in female runners with medically diagnosed injuries: a prospective investigation. *Br J Sports Med*. 2016;50(14):887-892.
11. Ferber R, McClay Davis I, Williams DS III. Gender differences in lower extremity mechanics during running. *Clin Biomech*. 2003;18(4):350-357.
12. Grood ES, Suntay WJ. A joint coordinate system for the clinical description of three-dimensional motions: application to the knee. *J Biomech Eng*. 1983;105(2):136-144.
13. James SL, Bates BT, Osternig LR. Injuries to runners. *Am J Sports Med*. 1978;6(2):40-50.
14. Khowailed IA, Petrofsky J, Lohman E, Daher N. Six weeks habituation of simulated barefoot running induces neuromuscular adaptations and changes in foot strike patterns in female runners. *Med Sci Monit*. 2015;21:2021-2030.
15. Milner CE, Ferber R, Pollard CD, Hamill J, Davis IS. Biomechanical factors associated with tibial stress fracture in female runners. *Med Sci Sports Exerc*. 2006;38(2):323-328.

16. Nigg BM. The role of impact forces and foot pronation: a new paradigm. *Clin J Sport Med*. 2001;11(1):2-9.
17. Nigg BM, Mohr MM, Nigg SR. Muscle tuning and preferred movement path—a paradigm shift. *Curr Issues Sport Sci*. 2017;2(84):1-12.
18. Perry SD, Radtke A, Goodwin CR. Influence of footwear midsole material hardness on dynamic balance control during unexpected gait termination. *Gait Posture*. 2007;25(1):94-98.
19. Pohl MB, Hamill J, Davis IS. Biomechanical and anatomic factors associated with a history of plantar fasciitis in female runners. *Clin J Sport Med*. 2009;19(5):372-376.
20. Pohl MB, Mullineaux DR, Milner CE, Hamill J, Davis IS. Biomechanical predictors of retrospective tibial stress fractures in runners. *J Biomech*. 2008;41(6):1160-1165.
21. Pollard CD, Ter Har JA, Hannigan JJ, Norcross MF. Influence of maximal running shoes on biomechanics before and after a 5k run. *Orthop J Sports Med*. 2018;6(6):2325967118775720.
22. Robbins S, Waked E, Allard P, McClaran J, Krouglicof N. Foot position awareness in younger and older men: the influence of footwear sole properties. *J Am Geriatr Soc*. 1997;45(1):61-66.
23. Robertson DGE, Dowling JJ. Design and responses of Butterworth and critically damped digital filters. *J Electromyogr Kinesiol*. 2003;13(6):569-573.
24. Saragiotto BT, Yamato TP, Hespanhol Junior LC, Rainbow MJ, Davis IS, Lopes AD. What are the main risk factors for running-related injuries? *Sports Med*. 2014;44(8):1153-1163.
25. Sinclair J, Richards J, Selfe J, Fau-Goodwin J, Shore H. The influence of minimalist and maximalist footwear on patellofemoral kinetics during running. *J Appl Biomech*. 2016;32(4):359-364.
26. Sinclair J, Richards J, Shore H. Effects of minimalist and maximalist footwear on Achilles tendon load in recreational runners. *Comp Exerc Physiol*. 2015;11(4):239-244.
27. Sobhani S, Hijmans J, van den Heuvel E, Zwerver J, Dekker R, Postema K. Biomechanics of slow running and walking with a rocker shoe. *Gait Posture*. 2013;38(4):998-1004.
28. Sobhani S, van den Heuvel ER, Dekker R, et al. Biomechanics of running with rocker shoes. *J Sci Med Sport*. 2017;20(1):38-44.
29. Taunton J, Ryan M, Clement D, McKenzie D, Lloyd-Smith D, Zumbo B. A retrospective case-control analysis of 2002 running injuries. *Br J Sports Med*. 2002;36(2):95-101.
30. Yamato TP, Saragiotto BT, Lopes AD. A consensus definition of running-related injury in recreational runners: a modified Delphi approach. *J Orthop Sports Phys Ther*. 2015;45(5):375-380.
31. Zadpoor AA, Nikooyan AA. The relationship between lower-extremity stress fractures and the ground reaction force: a systematic review. *Clin Biomech (Bristol, Avon)*. 2011;26(1):23-28.